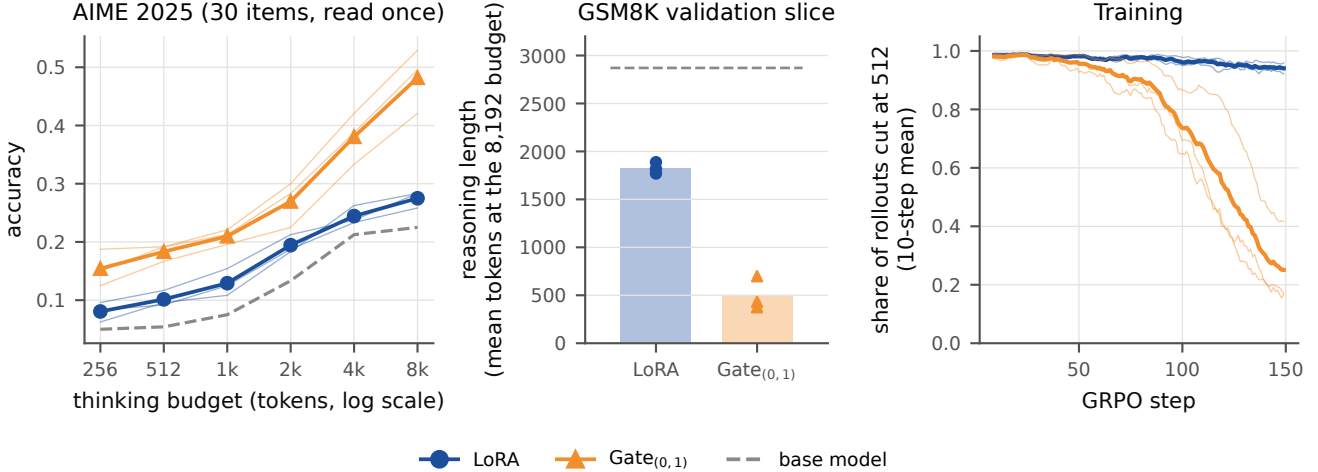
AIME 2025 (30 items, read once)
accuracy
0.5
0.4
0.3
0.2
0.1
256
512
1k
2k
4k
8k
thinking budget (tokens, log scale)
GSM8K validation slice
reasoning length
(mean tokens at the 8,192 budget)
3000
2500
2000
1500
1000
500
0
LoRA
Gate(0, 1)
Training
share of rollouts cut at 512
(10-step mean)
1.0
0.8
0.6
0.4
0.2
0.0
50
100
150
GRPO step
LoRA
Gate(0, 1)
base model